

AnalystLab Africa Experience Lab

FinTrust Digital Bank — Project Brief

Project: FinTrust Financial Intelligence & Digital Banking Support Solution

Project Duration: 4 Weeks **Project Type:** Multidisciplinary Data & AI Experience Lab

Organisation: FinTrust Digital Bank (fictional)

Important: FinTrust is a fictional organisation. All customers, transactions, labels and business information used in this project are synthetic and intended for educational purposes only.

1. Business Scenario

FinTrust Digital Bank is a growing digital banking organisation serving customers through mobile and online banking channels. As its customer base grows, FinTrust is collecting increasing amounts of customer and transaction data. However, the organisation needs a stronger way to turn this information into actionable business intelligence and reliable digital solutions.

Management wants to better understand customer behaviour, transaction patterns and potential financial-risk indicators, while also improving operational decision-making and digital customer support. FinTrust has therefore assembled a multidisciplinary team of Data Analysts, Data Scientists, ML Engineers, Generative AI practitioners and Project Managers.

2. Business Problem

FinTrust currently has customer and transaction information, but lacks an integrated solution that can:

- Provide clear visibility into customer and transaction behaviour.
- Identify potentially unusual or higher-risk transaction patterns.
- Generate predictive insights that can support risk-related decision-making.
- Operationalise analytical or machine-learning outputs into a usable workflow.
- Provide customers with reliable digital support based on approved FinTrust information.
- Bring these capabilities together into a coherent business solution.

3. Project Objective

Primary Objective

To develop an integrated data and AI solution that helps FinTrust Digital Bank understand customer and transaction behaviour, identify potential financial-risk patterns, support operational decision-making and improve digital customer support.

Supporting Objectives

- Analyse financial and customer data.
- Identify meaningful business trends and patterns.
- Develop relevant KPIs and business insights.
- Build and evaluate a predictive model for an appropriate risk-related problem.
- Develop an operational workflow around the model.
- Build a controlled GenAI customer-support capability.

- Apply appropriate testing and validation.
- Integrate outputs from multiple technical disciplines.
- Communicate findings and recommendations professionally.

4. Key Business Questions

Customer & Transaction Intelligence

- Who are FinTrust's major customer segments?
- How do transaction behaviours differ across customer groups?
- What transaction patterns are most important to management?
- Which products, channels or transaction types generate the greatest activity?

Risk & Predictive Intelligence

- What characteristics are associated with potentially risky transactions?
- Can historical transaction information be used to predict whether a transaction requires risk review?
- How well does the predictive model perform?
- Which factors contribute most to the model's predictions?

Customer Support & GenAI

- How can FinTrust provide faster answers to common customer questions?
- Can a GenAI assistant answer questions using only approved FinTrust information?
- How should unsupported or sensitive questions be handled?
- How can the assistant's responses be evaluated for accuracy and safety?

5. Project Scope

In Scope

- **Data Analytics:** data exploration, cleaning, KPI development, customer/transaction analysis, dashboarding and business insights.
- **Data Science:** exploratory analysis, feature engineering, predictive modelling, model evaluation, error analysis and interpretation.
- **ML Engineering:** model/pipeline integration, data/model workflow, API or service concepts, testing and technical documentation.
- **Generative AI:** prompt engineering, knowledge integration, customer-support assistant, RAG/knowledge-retrieval concepts, evaluation and guardrails.
- **Project Management:** planning, WBS, timeline management, dependency management, risk management, coordination, integration and final delivery.

Out of Scope

- Processing real customer banking information.
- Connecting to real bank accounts or executing real financial transactions.
- Providing personalised financial or investment advice.
- Making real lending, fraud or account-opening decisions.
- Deploying a production banking system.

- Making regulatory or legal claims beyond the approved fictional project resources.

6. Multidisciplinary Team Structure

Track	Primary Responsibility
Project Management	Planning, coordination, risks, dependencies and delivery.
Data Analytics	KPIs, dashboards, business analysis and insights.
Data Science	Predictive modelling and model evaluation.
ML Engineering	Model/pipeline operationalisation and technical integration.
Generative AI	Knowledge-based customer-support capability, evaluation and safety.

Team Principle: Each intern remains individually accountable for their own track deliverables. Formal cross-track collaboration begins in Week 2, after interns have produced tangible outputs.

7. Expected Final Solution

The team is expected to produce a coherent FinTrust Digital Banking Intelligence & Support Solution. The exact implementation may vary by track, but each component must have a clear purpose and relationship to the overall business problem.

Illustrative flow: Customer & Transaction Data → Analytics & KPIs → Predictive Risk Intelligence → ML Workflow/Service → Knowledge-Based GenAI Support → Integrated FinTrust Solution

8. Four-Week Project Journey

Week	Phase	Core Focus
Week 1	Understand & Plan	Business understanding, resource review, data/knowledge assessment, objectives and individual planning.
Week 2	Analyse & Prepare	Analysis, preparation, initial development and formal cross-track collaboration.
Week 3	Develop & Integrate	Advanced development, cross-track integration and validation.
Week 4	Test, Refine & Present	Testing, refinement, final integration and individual presentation.

9. Project Success Criteria

- **Business Understanding:** The solution addresses a clearly defined FinTrust business problem.
- **Data & Technical Quality:** Data, analysis, models and technical components are appropriately developed and documented.
- **Integration:** Outputs from different tracks are connected and actually used.
- **Validation:** The team tests its work and demonstrates improvements based on findings.
- **Responsible AI/Data Use:** The team recognises limitations, avoids unsupported claims and uses synthetic data appropriately.
- **Communication:** The solution can be explained clearly to technical and non-technical stakeholders.
- **Portfolio Readiness:** Work is documented and presented to a professional standard.

10. Data & AI Governance Principles

All interns must follow these principles throughout the project:

- Use only the supplied synthetic datasets and approved project resources unless an assignment explicitly permits another resource.
- Do not treat the synthetic Risk_Review_Flag as a real fraud determination.
- Do not provide personalised financial, investment or legal advice.
- The GenAI assistant must not invent FinTrust policies, fees, limits, timelines or procedures.
- Where approved information is unavailable, the assistant should acknowledge the limitation and follow the defined escalation approach.
- Document important assumptions, limitations and decisions.

11. Final Portfolio Expectations

Every intern will retain individual ownership of their track contribution. At the end of the project, each intern will submit an individual 5–10 minute recorded presentation explaining their contribution, how it connected to the wider solution, key results, limitations and lessons learned.

12. Approved Project Resources

Resource	Format	Purpose
FinTrust Project Brief	PDF	Business context, objectives, scope and success criteria
FinTrust 4-Week Master Roadmap	PDF	Four-week project and track progression
FinTrust Customer Data	CSV	Synthetic customer-level information
FinTrust Transaction Data	CSV	Synthetic transaction-level information
FinTrust Data Dictionary	Excel	Field definitions, business meaning and modelling guidance
FinTrust Financial Knowledge Base	Word	Approved knowledge source for the GenAI track

Document Purpose

This Project Brief serves as the master business reference for the four-week FinTrust Experience Lab. Weekly assignments should build on this document rather than introduce a separate business scenario.